BRAIN TEXT, BRAIN-TO-TEXT, AND THE MIND STYLE OF LITERATURE

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Abstract

Zhenzhao Nie's ethical literary criticism, specifically his theory of brain text and ethical selection, envisions an innovative and energetic understanding of the psychological and ethical processing of literature in the brain. This is evidenced by a survey of the ontological controversy between dualism and monism and a review of contemporary research in neuroscience, together with an investigation of the concept of brain text. Based on a comparative study of the neurolinguistic and neuroethical approaches and Nie's theory of brain text, the paper endorses Nie in his proposal of the brain text as the source of literature and his advocating of the teaching function of brain text (and literature). Meanwhile, the paper finds that the fruits of modern neuroscience contribute to Nie's stance: Brain text is a biological and material form with moral attributes. The idea of brain text coincides well with the findings of neuroscientists who have developed a system called brain-to-text that transforms brain activity into corresponding textual representation and have demonstrated that human behavior can be ethically energized in the strata of the brain. Cognitively, Nie's theory of brain text can be employed to decode the mind style of a writer, specifically a particular writer's ethical orientation as embodied in their choice of the order of words and syntax, which consciously and unconsciously represents the writer's world view.

Keywords

ethical literary criticism, neurotechnologies, mind-brain interaction, cognitive stylistics

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INTRODUCTION

Recent research in neuroscience has shown the possibility of "mind-reading," once a popular imaginationary power in science fiction, in 2015, real-world scientists subsequently made several important steps "toward human-machine communication based on imagined speech" by simply using a computer to read one's mind (Herff et al.) when scientists in Albany, New York, demonstrated for the first time the possibility of turning a person's thoughts into "a legible phrase using what they call a brain-to-text interface" (Koebler). The potential of this new technique is promising enough that some furturists believe that by 2045, humans will "achieve digital immortality by uploading their minds to computers" (Lewis).

However, two years before the neuroscientific concept of "brain-to-text" drew the attention of the public, Zhenzhao Nie in a fundamental article titled "Ethical Literary Criticism: Oral Literature and Brain Text," coined the term "Brain Text," namely, "a text stored in the human brain with a biological form, which preserves the memory of human perception and cognition and makes possible the production and circulation of oral literature before written language was invented" (11). In this paper, Nie further argues that "the human brain, as long as it lives and works, will never stop producing brain texts" (14). These brain texts occur in three major forms: oral, material, and electronic (digital). The theory of brain text may explain the origin of oral literature and shed new light on the understanding of the ethical function of literature, which will be illustrated later in this paper.

Anatomically, it is reasonable for us to accept the physical existence of human brain and its neurolinguistic functions, though there is no consensus concerning the neuroethical functions of the brain. However, it is likely that people might raise several doubts regarding scientific evidence suggesting the physical existence of brain text as previously defined: Is brain text a physical and biological form that is specifically localized in the brain and can be materially detected and measured (as brain waves) with scientific instruments? How far is brain text different from memory (and thoughts or mind)? How is brain text transferred into oral, written, and electronic (digital) texts? What is the philosophical and neuroscientific relationship among brain, mind, consciousness, and brain text? Besides, this paper raises an additional question: if we accept Nie's theory of brain text, how practicable and meaningful is it when employed to analyze the mind style of literature, both ethically and cognitively?

Despite the inevitable ethical controversies of mind-reading and mind-uploading of AI technology, increasingly advanced "brain-to-text" technological achievements in neuroscience, together with neuroscientist's new findings of consciousness as a quantum phenomenon enhance the reliability of Nie's theory and increase the possibility of brain text (and even by association, the related concept of consciousness) taking on a physical form. At the very least, Nie's theory of brain text provides an effective way to investigate a literary text's "mind style" (Fowler, 103), which linguistically presents an individual mental self and dramatizes the order and the structure of conscious (and unconscious) thoughts while displaying a character or narrator's perspectives and their moral and aesthetic bias. This will be further demonstrated in my case analysis.

FROM DUALISM TO MONISM: BRAIN, MIND, AND BRAIN-TO-TEXT

To understand the significance of brain text as a concept, one must first illustrate the gap between the physical organ of the brain and the mental concept of the mind or the consciousness, something which has been relevant in both neurological and philosophical fields and which the theory of brain texts intends to elucidate. The mind-body relation remains an important issue in neurophilosophy as this connection involves the eternal controversy of whether the brain and the mind (soul) are one and the same or are divided and separate. Two schools of thought have emerged in response to this problem: dualism and monism. The former proposes the existence of distinction of mind and brain (or body), while the latter regards all the different manifestations of the universe as a unified common entity.

The historical background of the mind-body connection can be traced to the ancient Greek period. Then, people understood that the brain instead of the heart was the source of the mind. During about 400 A.D., St. Augustine established much of the theological teaching of the early Catholic Church. As Kauffman and Radin explained, Augustine enriched the exploration of the mind as he posited that "human consciousness was due to a direct connection to the Mind of God" (Kauffman and Radin 2). In contrast, in 1640, Rene Descartes proposed the famous Cartesian dualism that, as Davies writes, regarded human bodies "as material things which are in essence extended in space," and human minds as "immaterial things whose essence is thinking" (Davies 133). The Cartesian dualism refuses to accept the mind as something housed within human body, because the mind does not possess spatial properties. Descartes's ontological dualism of mind and body made it difficult for him to describe the phenomenology of embodiment which is the way we experience our own body. However, when asked to elaborate on the phenomenology of embodiment, Descartes sometimes spoke of "the mind being 'mixed up with' the body" (Davies 133). He argued that "the mind could modulate processes in the material brain by way of a causal interaction operating through the pineal gland [in the brain]" (Davies 133). As Kauffman and Radin observed, this was perhaps "a first step to see the question of mind-brain interaction" (3), a mystery still unsolved today. My question, drawn from Descartes's stance, is: How do the brain and mind work upon each other neurolinguistically and neuroethically? This question could be a subject of further study in the fields of literary theory and neuroscience.

Something that complicates Descartes' dualism further is that the contemporary predominant theory of the mind-brain is materialism rather than idealism. Materialism postulates that all the universe is physical, and mind is no more than an expression of physical matter. According to materialism, consciousness spontaneously emerges out of the human neural network or the brain. This view is the theoretical basis for neuroscience (especially neurolinguistics and neuroethics) which takes "the neural correlates of consciousness" as its focus (Kauffman and Radin 2). However, the question of how the neural correlates deal with subjective experience and/or personal consciousness still remains a puzzle.

Although the materialist understanding of the world and human nature is dominant, Descartes's dualism has proven durable, since he has succeeded in bifurcating human personality into mind and body, a seminal and controversial issue that continues to this day. Despite of his unsatisfactory attempt at resolving the myth of brain-mind interaction, none of the theories after him can serve completely as alternative explanations to his account. The materialists claim that there are no "souls" separate from our physical selves and that our consciousness and physical selves must be seen as a unified totality; therefore, materialism tends to be more of a philosophy opposed to dualism and agnosticism in its epistemology. But the question is, if we reject the Cartesian conception of a divisible consciousness separate from the body and accept the materialist view of mind and consciousness determined by the matter, then to what extent is our consciousness (mind) determined by our brains? How does this account for human intuition and extrasensory perception?

Modern neuroscience has proven that brain signals can be measured and can reveal whether one is thinking about an animal or a non-living object. Our memory for word meaning is compartmentalized. When one thinks of a non-living object like a tool, a specific population of neural cells becomes active. In contrast, when one thinks of something living, such as a cat, that thought is processed by a different set of neurons. This is the discovery of neuroscientist Irina Simanova during her PhD studies at Radbound University Nijmegen "where she investigated the conceptual representation of words and objects in the human brain"; using EEG and fMRI, she was "the first to investigate whether the same neurons process different representations of one object–an image of a cat and the word 'cat'" (Radboud University Nijmegen). Simanova's examination of the neural networks offers insight into how we perceive objects, understand words, and produce language. Her work in identifying the image and its corresponding word processed in a different way in the human brain sheds some light on the mechanism of brain text, a key concept that ethical literary criticism takes as the source of all literatures.

Soon after Simanova's innovative work, neuroscientists made further contributions in communication between humans and machines based on natural speech related cortical activity. Over the past ten years, neuroscientists have made it possible to recognize isolated aspects (such as phones and isolated words) of speech from neural signals, but decoding continuous speech from the neural substrate in processing language remained an unsolved technological challenge.

Recently it was reported that neuroscientists successfully decoded spoken speech into the expressed words from intracranial electrocorticographic (ECoG) recordings for the first time. The scientists implemented a system called brain-to-text that models single phones, employs techniques from automatic speech recognition (ASR), and thereby transforms brain activity into corresponding textual representation. Christian Herff and Tanja Schultz used functional MRI and near infrared imaging "that can detect neural signals based on metabolic activity of neurons, to methods such as EEG and magnetoencephalography (MEG) that can detect electromagnetic activity of neurons responding to speech, successfully capturing neural signals from the brain and then decoding them to text" (Sundaresan). The brain-to-text system represents an important step toward human-machine communication based on imagined speech.

Language and the brain are closely interrelated and interactive. Human language is constantly changing and shaping the human brain, as the world-famous Chinese linguist and professor Wang Shiyuan asserts that "different languages shape different human brains" (138). One of the most essential features of the human brain is its capacity to produce language. Therefore, "to investigate the relation between language and brain is the best way to explore human nature, as human nature is a co-production of brain and language" (138).

Of all forms of languages, the language of literature is the liveliest representation of the working structure of mind and morality. In literary reading, the mechanism of our inner moral and aesthetic judgement remains a much more complicated neuroscientific issue. How is a piece of literary work (which is often regarded as a typical form of consciousness with aesthetic judgement) processed and produced? Ethical literary criticism proposes that the earliest form of literature is a verbal expression of brain text out of the need for conscious ethical communication.

FROM NEUROLINGUISTICS TO NEUROETHICS: BRAIN, BRAIN TEXT AND MORALITY

If one assumes that literature is above all a form of human communication, which makes creative use of language in order to provide entertainment, cognition, aesthetic judgement, and moral instruction, then it would be appropriate to investigate the human brain's language production. This leads to the discipline of neurolinguistics, which "studies the relation of language and communication to different aspects of brain function" (Ahlsén 3). The wide scope of neurolinguistics determines its interdisciplinary attribute, which combines neurology, linguistics, psychology and others. As Elisabeth Ahlsén rightly observes, neurolinguistics "has a very close relationship to psycholinguistics, but focuses more on studies of the brain" (3).

There are many theories relating the functions of the brain to its structures, especially its higher cognitive function and moral function. A materialist view of these functions would "follow the same patterns and laws as those of the 'simpler' motor and sensory functions" (Ahlsén 31) and search for the neurons cortically or subcortically localized in different areas of the brain. But this mechanical way of pure localism fails to describe the sophisticated and dynamic production of language. In her review of different neurolinguistic approaches to the functions of the brain, Elisabeth Ahlsén asserts that:

> Associationism and dynamic localization of function are more sophisticated views than pure localism, but still take basically the same approach. Holism, on the other hand, has a different theoretical background, where higher cognitive abilities are seen as different in nature from simpler ones and as, in principle, impossible to localize in specific cortical areas. These theories are therefore more dependent on psychology and less dependent on anatomy and physiology. (31)

In addition to neurolinguistics, neuroethics is another important approach to the study of the functions of the brain. *Stanford Encyclopedia of Philosophy* defines the term neuroethics as follows:

> Neuroethics is an interdisciplinary research area that focuses on ethical issues raised by our increased and constantly improving understanding of the brain and our ability to monitor and influence it, as well as on ethical issues that emerge from our concomitant deepening understanding of the biological bases of agency and ethical decision-making. (Roskies)

In other words, neuroethics encompasses two related fields of study, "the ethics of neuroscience" and "the neuroscience of ethics" (Roskies). Nowadays, increasingly new neurotechnologies are helping neuroscientists monitor and modify brain processes to change and modify human ethical behaviors, traits, or abilities, although there are no major breakthroughs in the field in the time of writing.

One of the tasks of neuroethics is studying moral identity and moral motivation from the perspective of neurobiology. Darcia Narvaez suggests that moral identity and moral motivation are not unitary constructs and argues that "instead humans have multiple moral motivations rooted in the evolved strata of the brain" (323). Considering this view, moral motivation shifts when a different mindset is active.

Narvaez posits that human behavior can be ethically energized in three forms in the strata of the brain: "Safety Ethic," "Engagement Ethic," and "Imagination Ethic" (323-340).

According to Narvaez, the first wave of brain evolution is the Safety Ethic rooted in the extrapyramidal action nervous system, which mostly relates to primitive instincts for survival and thriving in context. Because survival mechanisms are "hardwired into the brain, they are not easily damaged and can become the default mindset when social support is lacking and brain development is suboptimal" (324). Therefore, the Safety Ethic is part of an earlier stage of evolution.

The second wave of brain evolution is Engagement Ethic, which is "an epigenetic intuition central to mammalian functioning, the limbic system and related structures" (Narvaez 325). Research shows human infant's brain and body systems are dependent on experience, as the brain is socially wired and constructed in the early years: "Development may be conceptualized as the transformation of external into internal regulation" where the "progression represents an increase of complexity of the maturing brain systems that adaptively regulate the interaction between the developing organism and the social environment" (Narvaez 326).

The third evolution of brain is Imagination Ethic, the most important stage of brain evolution, referring to the neocortex and related thamalic structures that are "[primarily] focused on the external world, providing the capacity for problem solving and deliberative learning" (Narvaez 327). Most research in morality focuses on reasoning, which is central to the Ethic of Imagination and is a capability that could be achieved by cultivated deliberation and narrative. Narvaez makes it clear that:

> [The] frontal lobes provide the relay station between emotions and goals, planning and doing, coordinating systems from all parts of the brain. They maintain the sense of identity in cultural context through narrative self-explanation. The development of brain areas related to the Ethic of Imagination, like those related to the Engagement Ethic, require a nurturing environment. (327)

The frontal lobes, which are the source of our deliberative reasoning, are a critical stage in human evolution.

Narvaez's theory of the Ethic of Imagination is coincidentally an echo to the brain text theory proposed by Nie in 2013. Both stress the roles of cultivated deliberation and narrative in nurturing one's personality and both give highlights to the ethical function of narrative (literature) as well as the moral and cultural environment for the growth of children and adults.

To some extent, Narvaez's neurobiological theory of the evolution of the human brain echoes Karl Marx's explanation of the evolution of the brain and its functions. Marx's understanding of human development is the fruit of modern views of the evolution of homo sapiens. Based on evolution, "the hominids were bipedal before they developed large brains" (Graham-Leigh). Bipedalism then freed the hands for making and using tools and carrying out labor. This, to Marx, was the driver for the growth of the large hominid brains, as Friedrich Engels states in *The Dialectics of Nature*: "the hand is not only the organ of labour, it is also the product of labour" (174). This shows that human labor is a dialectical process that not merely makes man on a social level but also shapes themselves on a biological one, through the interplay between one's genes and environment. In Steven Rose's summary of Marx's opinions on the roles of brain in human development, he states that:

> By virtue of our large brains.... humans have created societies, invented technologies and cultures, and in doing so have changed themselves, their states of consciousness and, in effect, their genes. We are the inheritors of not merely the genes, but also the cultures and technologies of our forebears. (105)

Ethical literary criticism holds that human beings experienced the stage of "natural selection" at the beginning of their evolutionary development when the formation and maturity of the brain were of primary importance. According to Nie, the formation of the human brain is the biological prerequisite of generating brain text. The notion of brain text encompasses two aspects: the human brain and the text. Here, the term "brain text" does not refer to the visible and palpable existence of such texts as printed text and electronic text or digital text. It is instead the abstract existence of human cognition and thinking. In his groundbreaking book *Introduction to Ethical Literary Criticism* (2014), Nie defines brain text as "a special biological form . . . the perception and understanding of the world preserved by the human brain in the form of memory" (115). He also holds that in the earliest stage of natural selection, the brains of apes gradually evolved into the forms of human brains through constant labor. In this sense, labor creates human beings as well as human brain.

After millennia of evolution, human brains gradually developed the ethical consciousness of good and evil and thus was able to produce ethical brain text. This is crucial for humans to behave reasonably and ethically based on their Ethical Selection, a process that defines the nature of humanity and differentiates humans from animals. Unlike the brains of other animals, the brains of humans have the exclusive quality of ethical consciousness to distinguish goodness from evil, and therefore can generate brain texts endowed with reasonable and ethical orientation. Presumably, the brain of an animal might also produce brain text, but the brain text it produces does not possess any reasonable and moral sense or value. This is the fundamental difference between man and animal.

Brain Text is a core concept of the theory of ethical literary criticism, as it might shatter the traditional concept of what literature is and where literature comes from. For Nie, the text of literature is neither an expression of aesthetic ideology nor the art of language as conventionally accepted in the critical academia. It is instead the existence of a material form, as Nie expounded in his 2013 article. However, Nie's definition leaves us with a theoretical puzzle: If literature appears in material form, then what about oral literature? Does oral literature have a material form?

Obviously, this seems self-contradictory, as oral literature (e.g. epic) is of course one of the earliest tradition of world literature. To answer this question, Nie brilliantly proposed in 2013 that oral literature in fact consists of stories and songs that are stored in the brain and later recollected, a process which resembles the creation of what Nie calls "brain text." Later in 2017, Nie further revised and expanded the concept of brain text:

> What is brain text? It refers to the text stored in the human brain… Through the thinking process of perception, cognition and understanding, the brain is able to store the results of thinking as memory text, which constructs the brain text. Brain text is a special biological form with human brain as its carrier. People's perception and cognition of the objective things are first stored in the brain in the form of brain concepts, by the help of which thinking is processed and as a result of thinking, thoughts are obtained. Human thoughts are the result of the brain's processing of the objective or abstract things based on perception, cognition and understanding, all contributing to the formation of brain text as long as they are stored in the brain. ("The Forming Mechanism" 30)

The excerpt above explains the formation of brain text by clarifying how people acquire brain concepts through their perception and how people employ brain concepts to process thinking and finally shape their thoughts. Here, we find the most important element in the formation of brain text is the "brain concept." Ethical literary criticism aims to enable the reader with the capability of constructing ethical concepts in the brain (mind) by imputing (reading) literature of ethical values.

Obviously, as a theoretical and humanistic assumption, the theory of brain text differs fundamentally from neuroscience which attempts to localize and specify the moral and linguistic areas of the brain to provide clinical therapy or to modify immoral behavior. The theory of brain text, however, is related to neuroscience in that both deal with the solution to the ethical problems of human beings but in different ways. Although the theories of neuroethics could be partly borrowed to enhance our understanding of the mechanism of brain text, ethical literary criticism never makes any attempt to decipher and measure the mechanism of literary creation by testing or measuring the writer's brain, which ethical critics would take as mechanically and ethically problematic.

Ethical literary criticism serves the purpose of ethical interpretation of literature. To this end, in 2017, Nie specifically identified brain text as the source of human morality and underlined its essential value for humanity:

> Brain text is an established procedure to determine people's thoughts and behaviors, it serves not only to exchange and disseminate information, but also to determine people's consciousness, thinking, judgment, choice, action, emotion… The thoughts, choice and behavior of human beings, including moral cultivation and spiritual pursuit, are all determined by the brain text. Brain text regulates people's way of life and moral behavior, determines the moral existence of human beings and hence defines the essence of humanity. ("The Forming Mechanism" 33)

In the same article, Nie suggests that the best way to obtain brain text is by reading literary works, "since literary works, especially written texts and electronic texts, can be converted directly into brain texts through human visual or auditory organs" (33). Of course, this does not mean the brain text obtained by the reader is a passive receptacle or a secondary organism of the author's brain text. Ethical literary criticism proposes the reader's initiative in literary interpretation and encourages the reader's moral involvement and understanding of a literary work under special ethical and historical contexts. The ultimate purpose of ethical literary criticism is to guide people to examine their own ideas of morality and ethics by investigating the moral values of a piece of literary work.

The concept of brain text solidifies the theoretical basis of ethical literary criticism since it goes beyond the clinical and mechanical orientation of neuroscience to explore both ethical and aesthetic functions of literature. This could be exemplified in the following analysis of the mind style of literature as a reflex of brain text.

MIND STYLE OF LITERATURE AS A REFLEX OF BRAIN TEXT

Linguistically, the definition of the term "text" remains controversial. In his article "What is a Text? Explanation and Understanding," Paul Ricoeur suggests that "a text is any discourse fixed by writing" (145). In his definition of "text", Guy Cook elaborates Ricoeur's definition by regarding the text as "the linguistic forms in a stretch of language… Text is dependent on its receiver… and interacts with contexts [both internal and external]" (24). Then he moves further on to define the term "discourse" as something "opposed to text," "a coherent stretch of language in use, taking on meaning in context for its users, and perceived by them as purposeful, meaningful and connected" (25). This perceived quality of discourse is compatible with the mind style, which could be both a linguistic and moral reflex of brain text.

Leech and Short, in their co-authored book *Style in Fiction* (1981), devoted a whole chapter to illustrating the concept of "mind style." This concept was initiated by the linguist and literary critic Roger Fowler four years prior in *Linguistics and the Novel* and refers to "any distinctive linguistic representation of an individual mental self" (Fowler 103). Leech and Short believe that a writer's style may reveal "the particular writer's habitual way of experiencing and interpreting things" (151). They agree with Fowler's understanding of the innate relationship between the world perceived by a writer and the mind style of the writer: "Cumulatively, consistent structural options, agreeing in cutting the presented world to one pattern or another, give rise to an impression of a world-view, what I shall call a 'mind style'" (76). Mind style might also cognitively reveal the moral orientation of the brain text.

Take the following text (TEXT A) for example:

> This is just to say that I have eaten the plums that were the icebox. They were delicious, so sweet and so cold. But you were probably saving them for breakfast. Forgive me. (Widdowson 26)

If we take this as a natural daily note written on scrap of paper, left in the kitchen by a husband for a wife, it would mean no more than a little domestic confession and apology. The recipient of the note, the wife, would react with "exasperation, perhaps, or tolerant forbearance, and think no more about it" (Widdowson 27), since there is no reason to associate it with any moral or aesthetic intention.

If we rearrange the proposition of the note in TEXT A and fashion it into a little poem running as TEXT B, the visual structure and style would be different:

This Is Just to Say
I have eaten
the plums
that were in
the icebox

and which
you were probably
saving
for breakfast

Forgive me
they were delicious
so sweet
and so cold (Williams qtd. in Widdowson 27)

TEXT B is displayed as a longer note than TEXT A and looks different from TEXT A in structure and form.

In TEXT B, the reader is invited to experience and interpret the poem for a discovery of the value of its altering TEXT A's word and syntactical order. A closer scrutiny of the poem reveals that the expression of "Forgive me" is made prominent in TEXT B for its capital letter and its occurrence at the beginning of the last stanza of the little verse. Additionally, the first-person narrator (persona) of TEXT B emphasizes pleasure (rather than guilt) of tasting the plums by placing the word "delicious" as the end-focus of the poem. This choice of syntax structure and word order seems odd, as normally one would expect an apology to be followed by an expression of regret like TEXT A. "But here, on the contrary, what is expressed is not regret but relish. It is as if this first person is suggesting that the pleasure of experience is itself a reason for forgiveness" (Widdowson 29). Scholars of English literature might associate TEXT B to the biblical pleasure of tasting forbidden fruit, or perhaps relate it to the deprivation of "a long preserved virginity" in Andrew Marvel's famous poem "To His Coy Mistress" (Widdowson 28).

The secret pleasure conveyed in TEXT B is of course not without ethical and aesthetic values. Following ethical criticism, TEXT B betrays the first person's consciousness and subconsciousness in his brain text that is innate with ethical orientation. This explains the implication of the title of the poem which is distinct from the note. This is NOT just to say that he has eaten the plums in the icebox. "It is to say more than that, and it is in the manner of saying that this other dimension of meaning is represented" (Widdowson 30). To Widdowson's insightful comments, I would like to add that "this other dimension of meaning" is rooted in the ethical-oriented brain text.

If we reformulate TEXT B to the following version (TEXT C), we would be induced to read another different presentation of brain text. Suppose TEXT C reads like this:

This Is Just to Say
I have eaten the plums
that were in
the icebox

they were
delicious
so cold
so sweet

I'm sorry
you were probably
saving them
for breakfast (Widdowson 30)

TEXT C constitutes a different representation of the brain text and therefore a different interpretation of the mind style. TEXT C is a reshaping of TEXT A in the form of a poem but with little poetic implication and with few changes in TEXT A's the word or syntax order. We would of course be drawn into a quest for its representational significance, as it is presented as a poem. However, TEXT C presents a less poetic and philosophic interpretation compared with TEXT B, because by placing the three adjectives at the end of the poem TEXT B gives much more weight and end-focus to the pleasure of sensual experience of being "delicious, so sweet and so cold", and thus enhances the first-person speakers' desire for freedom and foregrounds an existentialist philosophy implied in the poem. In other words, the brain text of TEXT C would be almost a semantic duplicate of TEXT A. Its interpretation barely goes beyond TEXT A except in its form.

The analysis of TEXTS A, B, and C exhibits the moral significance of brain text and its correlative with mind style, which serves as a representation of both the "word-view" and "world view" of the writer in their word choice and syntax and, more significantly, as an indirect or metaphorical representation of the narrator's consciousness and subconsciousness rooted in his brain text.

In recent years, the notions of "mind" and "consciousness" have become increasingly central to the concerns of cognitive stylistics and cognitive narratology as well as neuroscience. Cognitive psychologists use the term "mind-reading" interchangeably with "theory of mind" to describe our innate ability to observe people's thoughts, feelings, beliefs, and desires, whether in verbal or non-verbal language. Lisa Zunshine observes that cognitive scientists "began to appreciate our mind-reading ability as a special cognitive endowment, structuring our everyday communication and cultural representations" (7). She points out that our ability to interpret the behavior of people in terms of their underlying states of mind "seems to be such an integral part of what we are as human beings that we could be understandably reluctant to dignify it with fancy terms and elevate it into a separate object of study" (7).

Therefore, when it comes to ethical discussions of literature, I suggest that we should not neglect the mind style of literature and its correlation with the brain text, which determines, directly or indirectly, the verbal and non-verbal expression of a speaker or narrator. The notion of mind covers almost every aspect of our inner activities, "namely not only just prototypically cognitive activities such as thinking and perceiving" (Semino 156), but also, according to Alan Palmer, "dispositions, feelings, beliefs and emotions" (19). Palmer's view of the mind is identical to Nie's concept of brain text: Palmer suggests that readers analyze the constructions of the characters' minds, while Nie advocates an exploration of the ethical sources of the characters' brain texts. The mind style of an individual writer is inextricably linked with the brain text wherein tone's personal moral and aesthetic values are well coordinated and regulated.

CONCLUSION

Nie's ethical literary criticism, specifically his theory of brain text and ethical selection, envisions an innovative and energetic understanding of the psychological and ethical processing of literature in the brain. This is evidenced by a survey of the ontological controversy between dualism and monism and a review of contemporary research in neuroscience, together with an investigation of the concept of brain text and its relevance towards literary analysis from an ethical perspective.

Based on a comparative study of the neurolinguistic and neuroethical approaches to the brain and Nie's theory of brain text, I agree more with Nie's proposal of taking brain text as the source of literature as well as his defense for the moral teaching function of literature. Meanwhile, I am glad to find that the fruits of modern neuroscience contribute to Nie's statement that brain text is a biological and material form with moral attributes. Cognitively, Nie's theory of brain text can be employed to decode the mind style of a writer, specifically a particular writer's ethical orientation as embodied in their choice of the order of words and syntax, which may either consciously and unconsciously represent or betray their world view.

While contemporary brain-to-text technology and the neuroscientists' envisioned possibility of mind-loading in the near future might have strengthened Nie's theory of brain text as a material form as well as the source of literature, they also pose new ethical questions for our understanding of a digitized (possibly AI-dominated) world. Nowadays, neuroscientists are making new discoveries of the possible physical property of consciousness, believing that "consciousness actually exists independently and outside of the brain as an inherent property of the universe itself like dark matter and dark energy or gravity" (Lazarus). I hope the theory of ethical literary criticism would in the future focus more on these complicated issues as consciousness, intuition, qualia, and free will and place them within the theoretical framework of brain text. I look forward to the development of Nie's theory as a practical solution to the ethical problems both in posthumanist fiction and in posthuman reality.

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